Generative Models: Recap

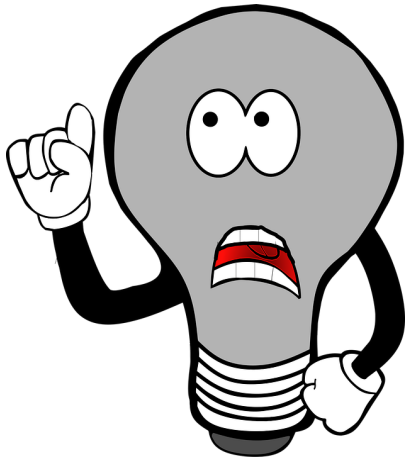
Goals

- Generate realistic samples
- Assign likelihood to samples (density Estimation)

Generative v.s. Discriminative Model

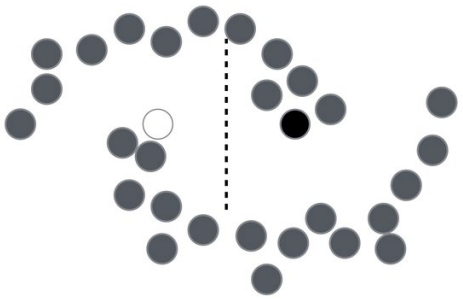
Goods and Bads

- Pros:
  - Out of distribution samples
  - Missing data
  - Missing dimensions
  - Semi-supervised learning
  - Synthetic sample generation
  - …
- Cons
  - Number of data points
  - Number of assumptions



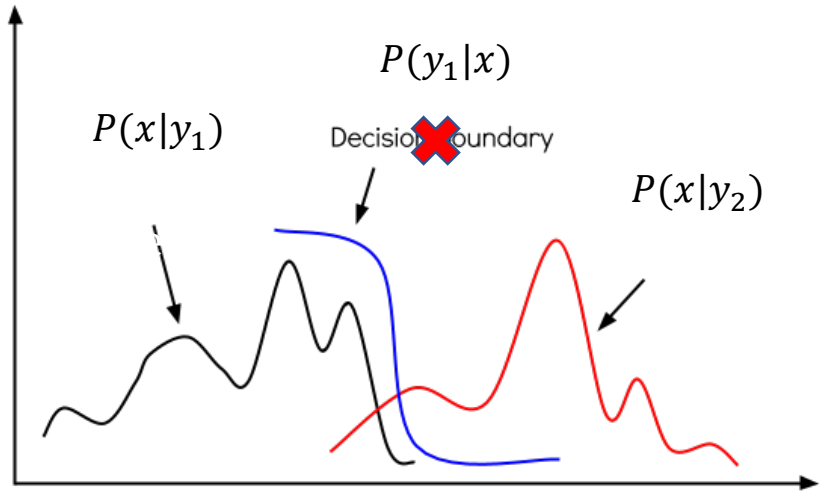








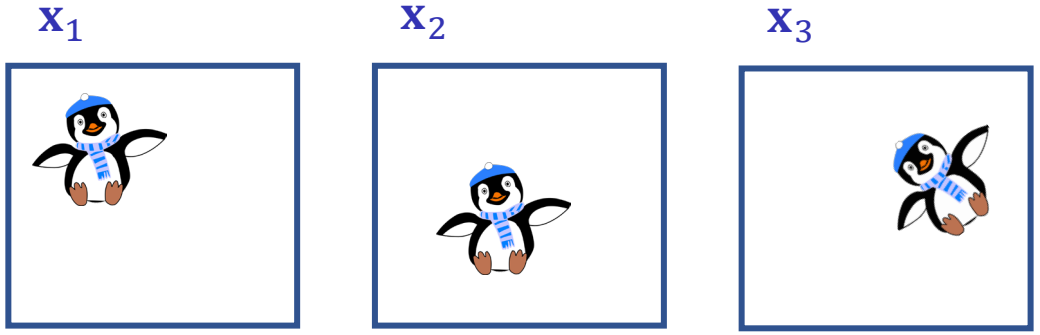
Despite the dimensionality of the input space ($\mathcal{X}$), why can we have a chance to train generative models in lack of enough data?



Generative Models: Goals

Generative Modeling Despite High Dimensionality of Input

- Underlying factors of variations, using simpler lower dimensional hidden variables, $\mathbf{z}$



$\mathbf{x} \in \mathbb{R}^{32\times32\times3}$

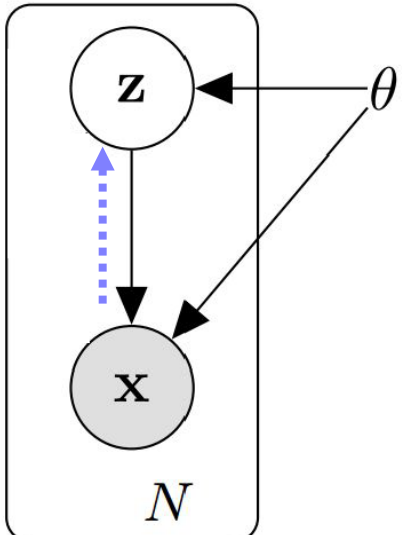
$z^{(1)}$: horizontal location
$z^{(2)}$: vertical location
$z^{(3)}$: rotation

$\mathbf{z} \in \mathbb{R}^{3}$

- 1st goal: $P_{\boldsymbol{\theta}}(\mathbf{z})$ and $P_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$

  and *maybe* $P_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$

- 2nd goal: $P_{\boldsymbol{\theta}}(\mathbf{x}) = P(\mathbf{x}|\boldsymbol{\theta}) = \int P(\mathbf{x}|\mathbf{z},\boldsymbol{\theta})P(\mathbf{z}|\boldsymbol{\theta})d\mathbf{z}$



Generative Models: Goals

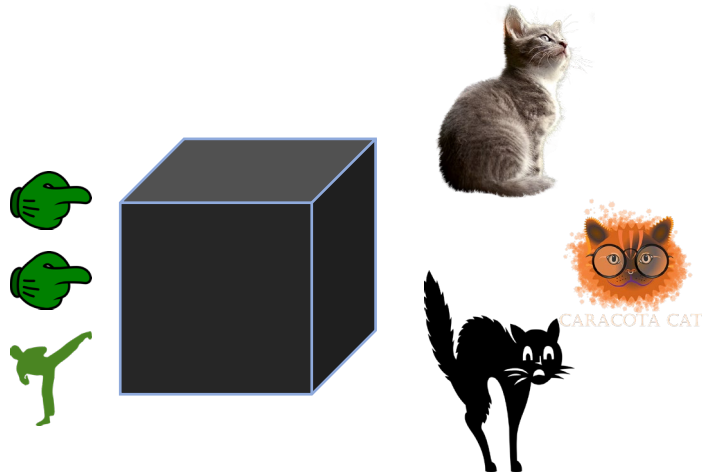
3rd Goal: Low-dimensional Representation Learning

- Generate realistic samples
- Assign likelihood to samples (Density Estimation)
- $P(\mathbf{z}|\mathbf{x})$: (Compressed) Representation Learning
  - Understanding underlying generative factors (e.g. similar to what PCA or matrix factorization do)
  - Data compression
  - Semi-supervised learning
  - ….

That was the probabilistic approach, do all models do that though?

Implicit Generative Models

- Not always, sometimes:
  - Discriminative model: $\underset{y}{\text{argmax}}\ score(\mathbf{x}, y)$
  - Generative model: $\mathrm{g}(\mathbf{z}) \rightarrow \hat{\mathbf{x}} \sim P(\mathbf{x})$ *or* $\hat{\mathbf{x}}, \hat{y} \sim P(\mathbf{x}, y)$

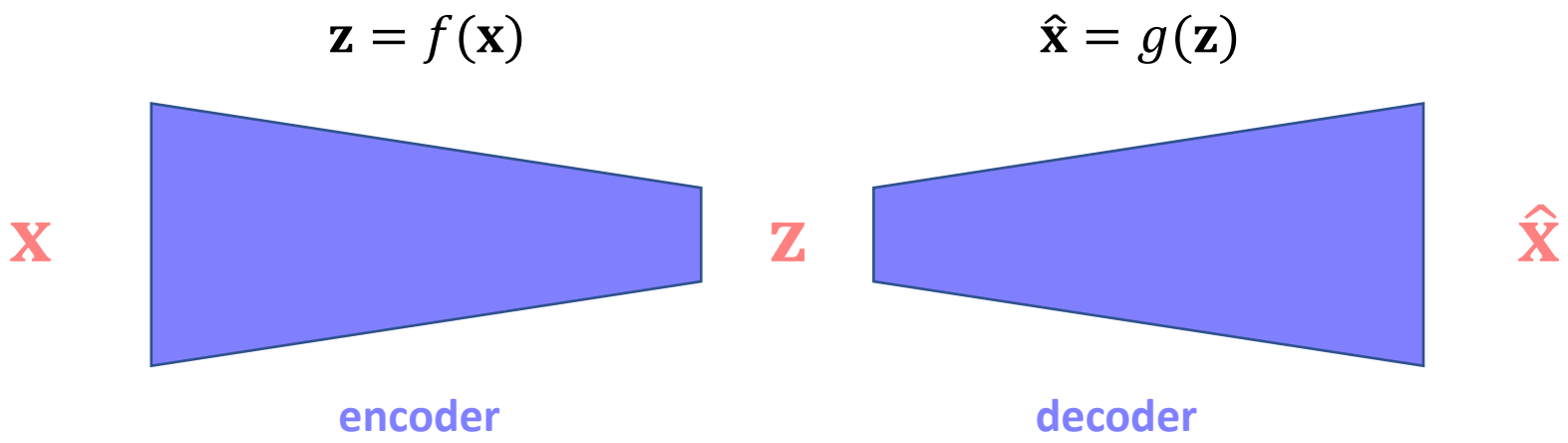


Deep Generative Models

Outline

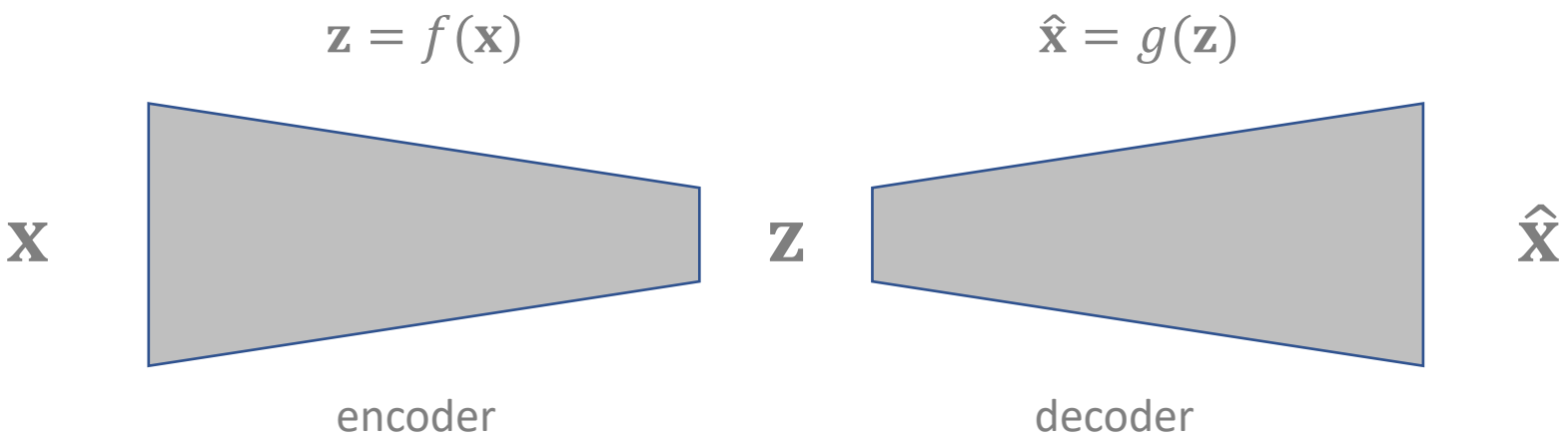
- Generative Modelling (Session 6)
- Variational Auto Encoders (Session 7)
  - Auto Encoders
  - Steps to create Variational Auto Encoders
  - Some Discussions
- Autoregressive Models (Session 8)
- Normalizing Flow Models (Session 9 or offline session)
- GANs (likely not covered)

(Bottleneck) Auto-Encoder Networks: Schematics



$\mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^d$ $\quad \mathbf{z} \in \mathbb{R}^l$ $\quad d \gg l$

(Bottleneck) Auto-Encoder Networks: Loss function

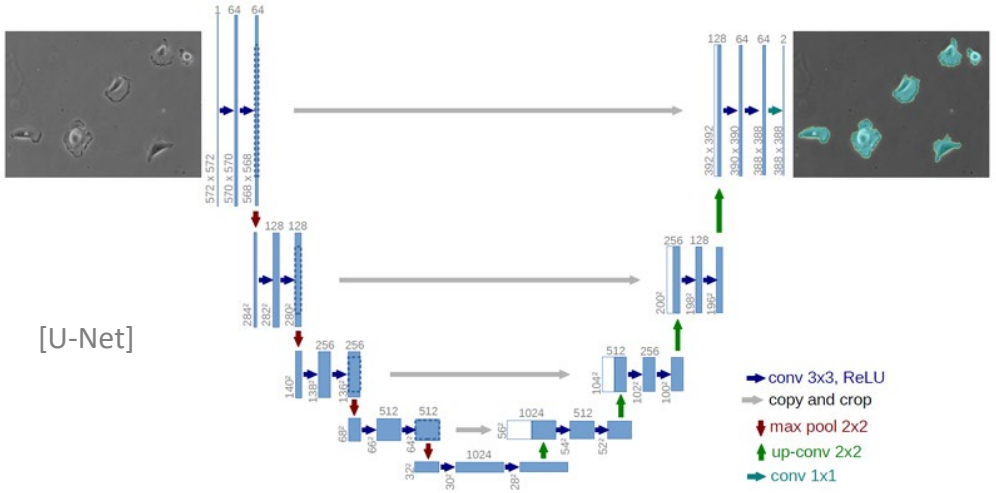


Loss function

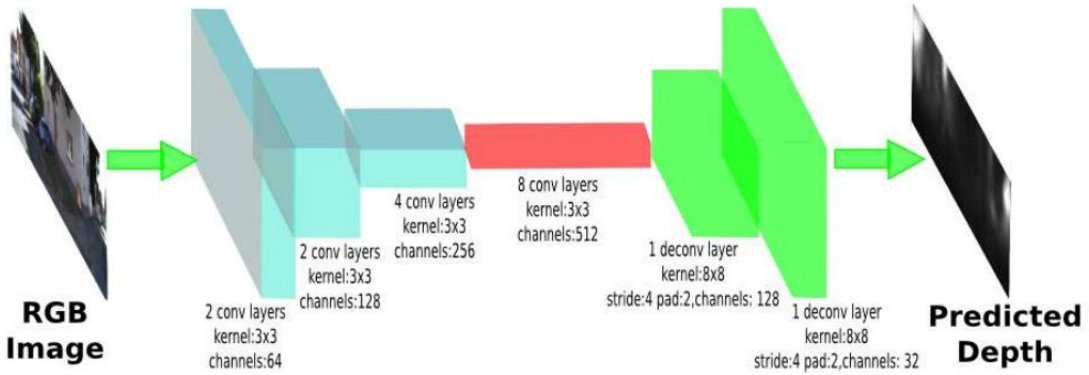
$$L = \sum_{i=1}^{n} ||\mathbf{x}_i - \hat{\mathbf{x}}_i||^2$$

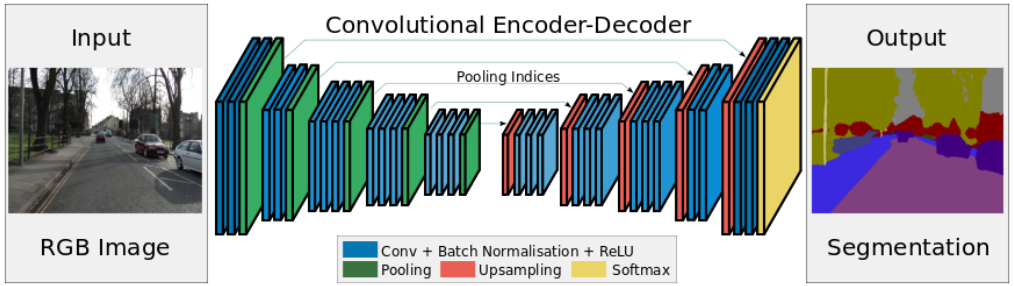
$\mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^d$ $\quad \mathbf{z} \in \mathbb{R}^l$ $\quad d \gg l$

(Bottleneck) Auto-Encoder Networks: similar discriminative architectures



[U-Net]

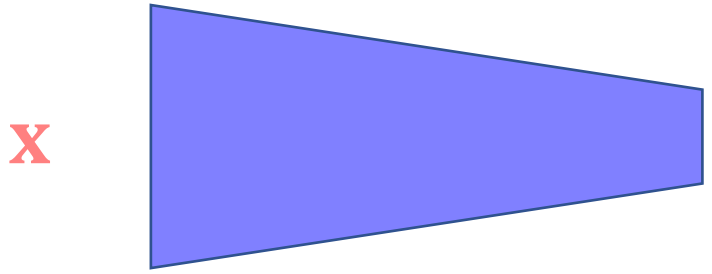




[SegNet]

(Bottleneck) Auto-Encoder Networks: use-cases

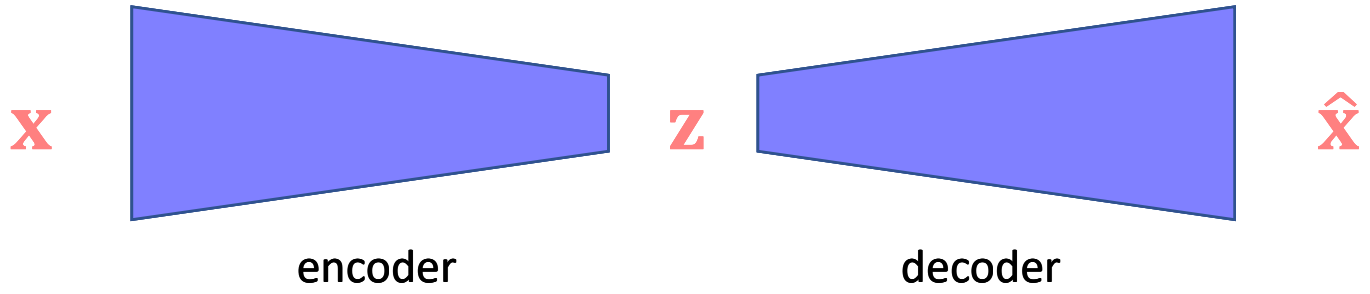
- Dimensionality Reduction
- Pretraining
- Denoising AutoEncoder
- Item imputation

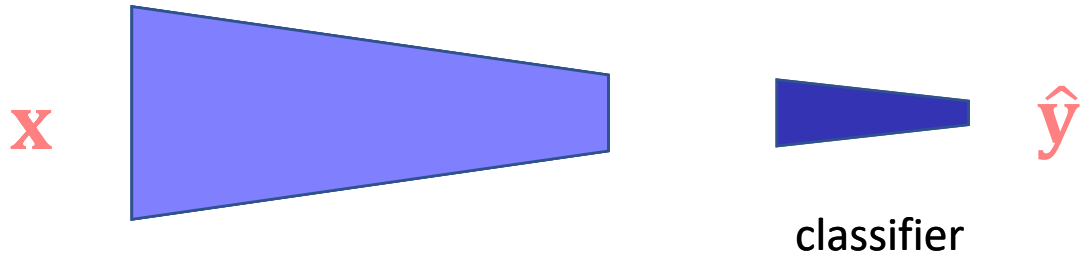


$\mathbf{x} \in \mathbb{R}^d$ $\mathbf{z} \in \mathbb{R}^p$ $d \gg p$

(Bottleneck) Auto-Encoder Networks: use-cases

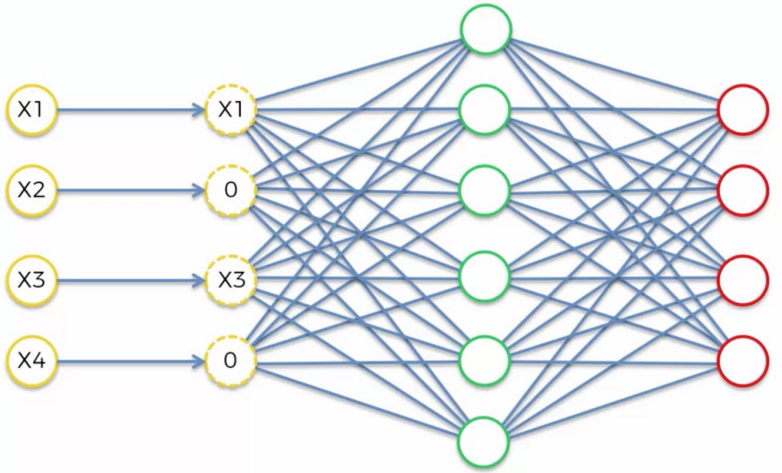
- Dimensionality Reduction
- Pretraining
- Denoising AutoEncoder
- Item imputation





(Bottleneck) Auto-Encoder Networks: use-cas

- Dimensionality Reduction
- Pretraining
- Denoising AutoEncoder
- Item imputation



[source: Kirill Eremenko]

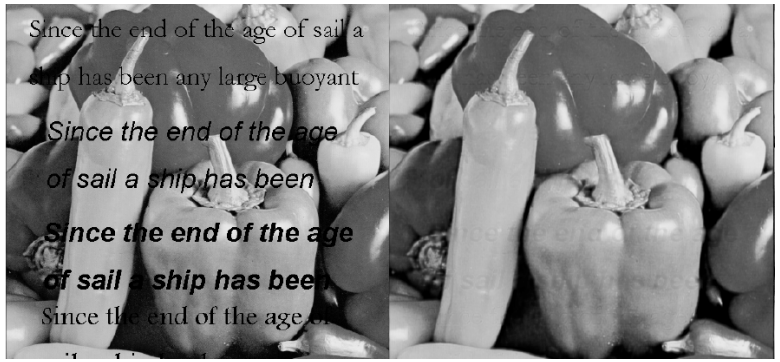


Vincent, H. Larochelle Y. Bengio and P.A. Manzagol, "**Extracting and Composing Robust Features with Denoising Autoencoders**", ICML 2008

(Bottleneck) Auto-Encoder Networks: use-cases

Dimensionality Reduction

- Pretraining
- Denoising AutoEncoder
- Item imputation (e.g. Image Inpainting)



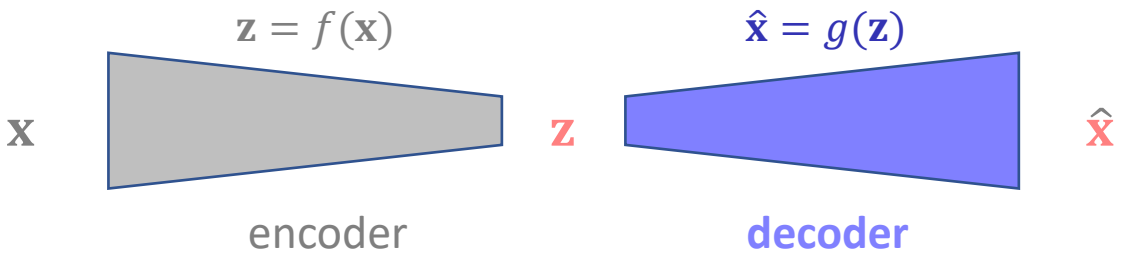
Reflection/Discussion point

- Are AutoEncoders generative models as we defined them?
- Why would we call it a generative model?
- Why would we not call it a generative model?



3 minutes!

Auto-Encoder Networks: generative models? not in principle!



Bengio, Yao, Alain, Vincent, "**Generalized Denoising Auto-Encoders as Generative Models**", NIPS 2013

Let’s make it probabilistic and truly generative!

- there are several steps!

Kingma, Welling "**Auto-Encoding Variational Bayes“,** ICLR 2013

VAE: implementation perspective

